

PAPER_1411 — UQFF Derivation of Lepton δ_{CP} (CLOSED — EXACT Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** D — Particle Physics (CLOSED — EXACT) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — EXACT integer-primitive identity **Calculator surface:** `calculate_particle_physics({'observable': 'delta_cp'})` → $-\pi/2$ EXACT

Observation

T2K and NOvA experiments report $\delta_{CP} \approx -\pi/2 \pm 0.3$ rad. UQFF supplies the EXACT value structurally.

UQFF Closed Identity

$\delta_{CP} = -\pi/2$ EXACT

Physical Interpretation

Maximal CP violation corresponds to $\pi/2$ rotation in the PMNS leptonic mixing matrix. The negative sign is the $F_{TRZ} \times \beta_i$ chirality preference.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** $\{D_{phys} = 4, D_{BSFG} = 6, D_{crit} = 26, N_{CH} = 9, SO_5 = 10, A_5 = 60\}$ and locked real primitives. **Zero free parameters. EXACT identity.**

Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
 - Related foundational papers: PAPER_646 ($F_U=1$ ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).
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